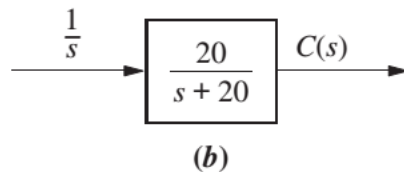
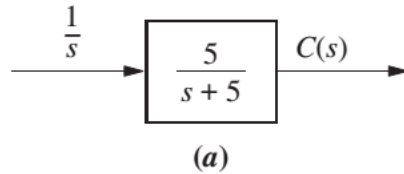


Assignment 9 (ELEC 341 L9_StepResponse)

Problem 1:

Find the output response, $c(t)$, for each of the systems shown in Figure. Also find the time constant, rise time, and settling time for each case.



FIGURE

$$T_s (2\%) = 4T$$

Use the difference between 10% to 90% of y_{ss} as the rise time and also use a 2% settling time.

a) $C(s) = H(s)R(s)$ $H(s) = \frac{5}{s+5} \Rightarrow T = \frac{1}{5} = 0.2s$

$$C(s) = \frac{5}{s+5} \cdot \frac{1}{s}$$

$$C(s) = \frac{5}{s(s+5)} \Rightarrow \int_0^t 5e^{-5\tau} d\tau = -e^{-5\tau} \Big|_0^t$$

$$c(t) = (1 - e^{-5t}) \cdot 1$$

$$T_s (2\%) = 4T$$

$$t_s (2\%) = 0.8s$$

$$c(t) = 1 - e^{-5t}$$

$$\ln(c(t)) = -5t$$

$$t = \frac{\ln(c(t))}{-5}$$

$$C_{ss} = \lim_{t \rightarrow \infty} c(t) = 1$$

$$10\% C_{ss} = 0.1$$

$$90\% C_{ss} = 0.9$$

$$t_r = \frac{1}{5} (\ln(0.9) - \ln(0.1))$$

$$t_r = 0.4394s$$

$$b). C(s) = H(s)R(s)$$

$$H(s) = \frac{20}{s+20}, \quad T = \frac{1}{20} \text{ s}$$

$$C(s) = \frac{20}{s+20} \cdot \frac{1}{s}$$

$$= \frac{20}{s(s+20)} \Rightarrow \int_0^t 20e^{-20\tau} d\tau = 20 \cdot \frac{1}{20} e^{-20\tau} \Big|_0^t$$

$$c(t) = (1 - e^{-20t}) \cdot 1(t)$$

$$c(t) = 1 - e^{-20t}$$

$$t_s (2\%) = 4\tau = \frac{1}{5}$$

$$c_{ss} = \lim_{t \rightarrow \infty} c(t) = 1.$$

$$t_s = 0.2 \text{ s}$$

$$10\% c_{ss} = 0.1$$

$$90\% c_{ss} = 0.9$$

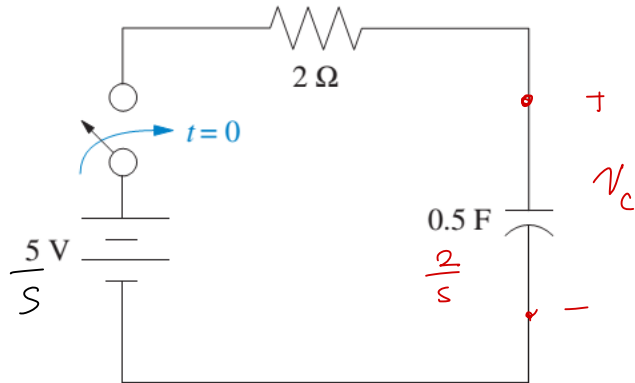
$$t_n = \frac{1}{20} \ln(9)$$

$$t_n = 0.1099 \text{ s}$$

Assignment 9 (ELEC 341 L9_StepResponse)

Problem 2:

Find the capacitor voltage in the network shown in Figure if the switch closes at $t = 0$. Assume zero initial conditions. Also find the time constant, rise time, and settling time for the capacitor voltage.



FIGURE

Use the difference between 10% to 90% of y_{ss} as the rise time and also use a 2% settling time. Use Matlab to verify and plot your output response.

$$V_c(s) = V_c(s) \cdot \frac{2/s}{2/s + 2} \Rightarrow H(s) = \frac{1}{s+1}$$

$$V_c(s) = \frac{5}{s} \cdot \frac{2/s}{2/s + 2}$$

$$= \frac{5}{s} \cdot \frac{2}{2s+2}$$

$$V_c(s) = \frac{5}{s} \cdot \frac{1}{s+1} \rightarrow \int e^{-\tau} d\tau = 1 - e^{-t}$$

$$V_c(t) = 5(1 - e^{-t})u(t)$$

$$t_s(2\%) = 4\tau$$

$$t_s(2\%) = 4s$$

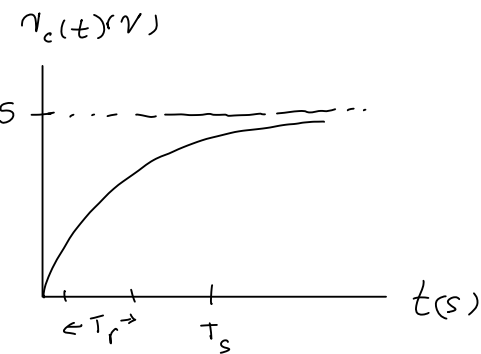
$$v_{ss} = \lim_{t \rightarrow \infty} v(t) = 5V$$

$$\ln\left(\frac{v_c(t)}{5}\right) = -t$$

$$t_r = \ln\left(\frac{0.9(5)}{5}\right) - \ln\left(\frac{0.1(5)}{5}\right)$$

$$t_r = \ln(9)$$

$$t_r = 2.1972s$$



Assignment 9 (ELEC 341 L9_StepResponse)

Problem 3:

For the second-order system that follows, find ζ , ω_n , T_s , T_p , T_r , and %OS.

$$T(s) = \frac{16}{s^2 + 3s + 16} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad T = \frac{1}{\zeta\omega_n} = \frac{1}{\frac{3}{8} \cdot 4} = \frac{2}{3} s.$$

$$\begin{aligned} \omega_n^2 &= 16 \\ \omega_n &= 4 \text{ rad/s} \\ 2\zeta\omega_n &= 3 \\ \zeta &= \frac{3}{2\omega_n} \\ \zeta &= \frac{3}{8} \end{aligned}$$

$$T_s (\%) = 4T = 8/3 s$$

$$T_p = \frac{\pi}{\omega_d}$$

$$T_p = \frac{2\pi}{\sqrt{55}} s$$

$$\begin{aligned} \omega_d &= \omega_n \sqrt{1 - \zeta^2} \\ &= 4 \sqrt{1 - \frac{9}{64}} \end{aligned}$$

$$\omega_d = 4 \frac{\sqrt{55}}{8}$$

$$\omega_d = \frac{\sqrt{55}}{2} \text{ rad/s}$$

$$\%OS = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \quad \zeta = 3/8$$

$$\%OS = 28.0597\%$$

$$T_r = \frac{1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1}{\omega_n}$$

$$T_r = 0.3559 s$$